DO NOW · AO1 REVISION

Stopping distance

Forces
Paper 2 · Combined and Separate
14 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	**What is the stopping distance of a vehicle?**	Thinking distance + braking distance	☐
2 ★	**What is the thinking distance of a vehicle?**	The distance travelled during the driver's reaction time.	☐
3 ★	**What is the braking distance of a vehicle?**	The distance travelled under the braking force.	☐
4	**Name the factor that increases both thinking distance and braking distance.**	Speed	☐
5	**State the typical range of human reaction times.**	0.2 s to 0.9 s	☐
6 ★	**Give four factors that affect reaction time.**	Tiredness, drugs, alcohol and distractions	☐
7 ★	**Give four factors that affect braking distance.**	Speed, wet or icy roads, worn tyres and worn brakes	☐
8 ★	**Describe the energy transfer when a vehicle brakes.**	Energy is transferred from the kinetic store of the vehicle to the thermal store of the brakes.	☐
9	**Describe how the braking force needed to stop in a given distance depends on speed.**	The greater the speed, the greater the braking force needed.	☐
10 ★	**Give two dangers of a very large deceleration.**	The brakes may overheat and the driver may lose control.	☐
11	**Describe a simple method to measure a person's reaction time.**	Drop a ruler between their fingers. The further it falls before they catch it, the longer the reaction time.	☐
12	**Explain why braking distance increases with speed.**	The vehicle has more kinetic energy, so the brakes must do more work to stop it.	☐

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	QUESTION	ANSWER	✓
13	**The speed of a vehicle doubles. State what happens to the thinking distance.**	It doubles (it is directly proportional to speed).	☐
14	**Which does speed affect more: thinking distance or braking distance?**	Braking distance	☐

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